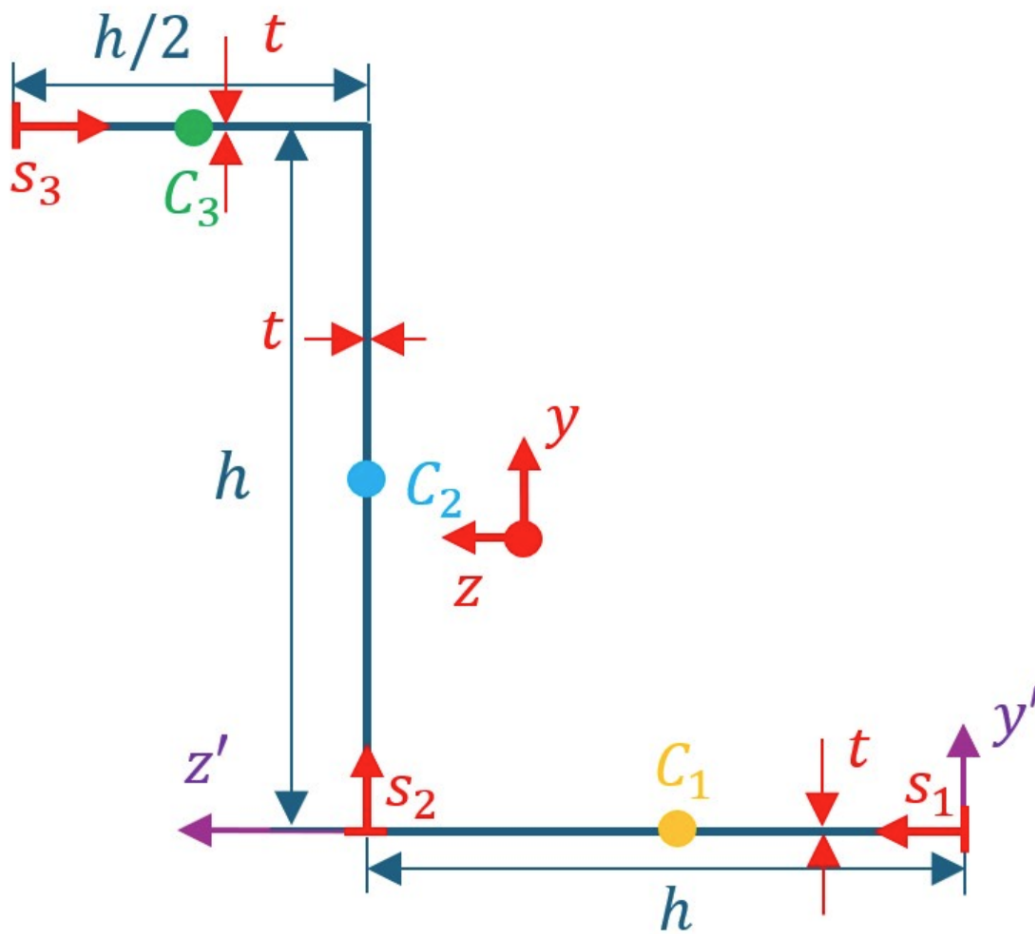


Thin-Wall 9

Image no stringer



No stringer

```
In[*]:= ClearAll["Global`*"]
(*Find Centroid Prime Axis*)
A1 = h * t;
A2 = t * h;
A3 =  $\frac{h}{2}$  * t;
Izz1 = 0;
Iyy1 =  $\frac{t * h^3}{12}$ ;
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Iyz1 = 0;
Izz2 =  $\frac{t * h^3}{12}$ ;
Iyy2 = 0;
Iyz2 = 0;
Izz3 = 0;
Iyy3 =  $\frac{t * (h / 2)^3}{12}$ ;
Iyz3 = 0;
ycp1 = 0;
zcp1 =  $\frac{h}{2}$ ;
ycp2 =  $\frac{h}{2}$ ;
zcp2 = h;
ycp3 = h;
zcp3 =  $\frac{5 * h}{4}$ ;
ycp =  $\frac{A1 * ycp1 + A2 * ycp2 + A3 * ycp3}{A1 + A2 + A3}$ ;
zcp =  $\frac{A1 * zcp1 + A2 * zcp2 + A3 * zcp3}{A1 + A2 + A3}$ ;

(*Translate to Global Coordinate*)
yc1 = ycp1 - ycp;
zc1 = zcp1 - zcp;
yc2 = ycp2 - ycp;
zc2 = zcp2 - zcp;
yc3 = ycp3 - ycp;
zc3 = zcp3 - zcp;

(*Find Second Area Moments*)
Iyy = Iyy1 + zc12 * A1 + Iyy2 + zc22 * A2 + Iyy3 + zc32 * A3;
Izz = Izz1 + yc12 * A1 + Izz2 + yc22 * A2 + Izz3 + yc32 * A3;
Iyz = Iyz1 + yc1 * zc1 * A1 + Iyz2 + yc2 * zc2 * A2 + Iyz3 + yc3 * zc3 * A3;

(*Parameterize 3 branches:*)
(*Prime coordinates for starting and final points for each branch*)
yps1i = 0;
yps1f = 0;
zps1i = 0;
zps1f = h;
yps2i = 0;
yps2f = h;
zps2i = h;

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```

zps2f = h;
yps3i = h;
yps3f = h;
zps3i =  $\frac{3 * h}{2}$ ;
zps3f = h;
(*Centroid coordinates for starting and final*)
ys1i = yps1i - ycp;
ys1f = yps1f - ycp;
zs1i = zps1i - zcp;
zs1f = zps1f - zcp;
ys2i = yps2i - ycp;
ys2f = yps2f - ycp;
zs2i = zps2i - zcp;
zs2f = zps2f - zcp;
ys3i = yps3i - ycp;
ys3f = yps3f - ycp;
zs3i = zps3i - zcp;
zs3f = zps3f - zcp;
(*Distances*)
ds1 = Simplify[ $\sqrt{(ys1f - ys1i)^2 + (zs1f - zs1i)^2}$ ];
ds2 = Simplify[ $\sqrt{(ys2f - ys2i)^2 + (zs2f - zs2i)^2}$ ];
ds3 = Simplify[ $\sqrt{(ys3f - ys3i)^2 + (zs3f - zs3i)^2}$ ];
(*Integrand Function*)
y1s1[s1_] =  $\frac{ys1f - ys1i}{ds1} * s1 + ys1i$ ;
z1s1[s1_] =  $\frac{zs1f - zs1i}{ds1} * s1 + zs1i$ ;
y2s2[s2_] =  $\frac{ys2f - ys2i}{ds2} * s2 + ys2i$ ;
z2s2[s2_] =  $\frac{zs2f - zs2i}{ds2} * s2 + zs2i$ ;
y3s3[s3_] =  $\frac{ys3f - ys3i}{ds3} * s3 + ys3i$ ;
z3s3[s3_] =  $\frac{zs3f - zs3i}{ds3} * s3 + zs3i$ ;

(*First Area Moment at Sbar*)
Qz1[s1b_] = t * Integrate[y1s1[s1], {s1, 0, s1b}];
Qy1[s1b_] = t * Integrate[z1s1[s1], {s1, 0, s1b}];
Qz2[s2b_] = t * Integrate[y2s2[s2], {s2, 0, s2b}];
Qy2[s2b_] = t * Integrate[z2s2[s2], {s2, 0, s2b}];
Qz3[s3b_] = t * Integrate[y3s3[s3], {s3, 0, s3b}];
Qy3[s3b_] = t * Integrate[z3s3[s3], {s3, 0, s3b}];

```

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(*Bottom Right S1*)
qs1vy0 = 0;
qs1vz0 = 0;
qs1vy[s1b_] =  $\frac{V_y * I_{yz}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{y1}[s1b] - \frac{V_y * I_{yy}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{z1}[s1b] + qs1vy0 // \text{Simplify};$ 
qs1vz[s1b_] =  $-\frac{V_z * I_{zz}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{y1}[s1b] + \frac{V_z * I_{yz}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{z1}[s1b] + qs1vz0 // \text{Simplify};$ 
(*Middle Section S2*)
qs2vy0 = qs1vy[h];
qs2vz0 = qs1vz[h];
qs2vy[s2b_] =  $\frac{V_y * I_{yz}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{y2}[s2b] - \frac{V_y * I_{yy}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{z2}[s2b] + qs2vy0 // \text{Simplify};$ 
qs2vz[s2b_] =  $-\frac{V_z * I_{zz}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{y2}[s2b] + \frac{V_z * I_{yz}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{z2}[s2b] + qs2vz0 // \text{Simplify};$ 
(*Top Left S3*)
qs3vy0 = 0;
qs3vz0 = 0;
qs3vy[s3b_] =  $\frac{V_y * I_{yz}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{y3}[s3b] - \frac{V_y * I_{yy}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{z3}[s3b] + qs3vy0 // \text{Simplify};$ 
qs3vz[s3b_] =  $-\frac{V_z * I_{zz}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{y3}[s3b] + \frac{V_z * I_{yz}}{I_{yy} * I_{zz} - I_{yz}^2} * Q_{z3}[s3b] + qs3vz0 // \text{Simplify};$ 

(*Find Branch Forces*)
F1vy = Integrate[qs1vy[s1b], {s1b, 0, h}];
F1vz = Integrate[qs1vz[s1b], {s1b, 0, h}];
F2vy = Integrate[qs2vy[s2b], {s2b, 0, h}];
F2vz = Integrate[qs2vz[s2b], {s2b, 0, h}];
F3vy = Integrate[qs3vy[s3b], {s3b, 0,  $\frac{h}{2}$ }];
F3vz = Integrate[qs3vz[s3b], {s3b, 0,  $\frac{h}{2}$ }];

(*Find Shear Center*)
Mxscvy = -F1vy * (ey - yc1) + F2vy * (ez - zc2) + F3vy * (ey - yc3);
Mxscvz = -F1vz * (ey - yc1) + F2vz * (ez - zc2) + F3vz * (ey - yc3);
eqneyez = Solve[{Mxscvy == 0, Mxscvz == 0}, {ey, ez}];
eys = ey /. eqneyez[[1]];
ezs = ez /. eqneyez[[1]];
ypey = eys + ycp;
zpey = ezs + zcp;

(*Questions 1-6*)
Q1 = qs1vy[0.0136] /. {Vy -> 23, Vz -> 0, h -> 0.0828, t -> 1.39 / 1000}
Q2 = qs2vy[0.0113] /. {Vy -> 23, Vz -> 0, h -> 0.0757, t -> 1.02 / 1000}

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A3 =  $\frac{h}{2} * t$ ;
As1 = 0.2 * h * t;
As2 = 0.2 * h * t;
Izz1 = 0;
Iyy1 =  $\frac{t * h^3}{12}$ ;
Iyz1 = 0;
Izz2 =  $\frac{t * h^3}{12}$ ;
Iyy2 = 0;
Iyz2 = 0;
Izz3 = 0;
Iyy3 =  $\frac{t * (h / 2)^3}{12}$ ;
Iyz3 = 0;
ycp1 = 0;
zcp1 =  $\frac{h}{2}$ ;
ycp2 =  $\frac{h}{2}$ ;
zcp2 = h;
ycp3 = h;
zcp3 =  $\frac{5 * h}{4}$ ;
yps1 = h;
zps1 = h;
yps2 = 0;
zps2 = h;
ycp =  $\frac{(A1 * ycp1 + A2 * ycp2 + A3 * ycp3) + (yps1 * As1 + yps2 * As2)}{(A1 + A2 + A3) + (As1 + As2)}$ ;
zcp =  $\frac{(A1 * zcp1 + A2 * zcp2 + A3 * zcp3) + (zps1 * As1 + zps2 * As2)}{(A1 + A2 + A3) + (As1 + As2)}$ ;

(*Translate to Global Coordinate*)
yc1 = ycp1 - ycp;
zc1 = zcp1 - zcp;
yc2 = ycp2 - ycp;
zc2 = zcp2 - zcp;
yc3 = ycp3 - ycp;
zc3 = zcp3 - zcp;
ys1 = (yps1 - ycp); (*Stringer*)
zs1 = (zps1 - zcp);
ys2 = (yps2 - ycp);
zs2 = (zps2 - zcp);

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(*Find Second Area Moments*)
Iyy = (Iyy1 + zc12 * A1 + Iyy2 + zc22 * A2 + Iyy3 + zc32 * A3) + (zs12 * As1 + zs22 * As2) // Simplify;
Izz = (Izz1 + yc12 * A1 + Izz2 + yc22 * A2 + Izz3 + yc32 * A3) + (ys12 * As1 + ys22 * As2) // Simplify;
Iyz = (Iyz1 + yc1 * zc1 * A1 + Iyz2 + yc2 * zc2 * A2 + Iyz3 + yc3 * zc3 * A3) +
      (zs1 * ys1 * As1 + zs2 * ys2 * As2) // Simplify;

(*Parameterize 3 branches:*)
(*Prime coordinates for starting and final points for each branch*)
yps1i = 0;
yps1f = 0;
zps1i = 0;
zps1f = h;
yps2i = 0;
yps2f = h;
zps2i = h;
zps2f = h;
yps3i = h;
yps3f = h;
zps3i =  $\frac{3 * h}{2}$ ;
zps3f = h;
(*Centroid coordinates for starting and final*)
ys1i = yps1i - ycp;
ys1f = yps1f - ycp;
zs1i = zps1i - zcp;
zs1f = zps1f - zcp;
ys2i = yps2i - ycp;
ys2f = yps2f - ycp;
zs2i = zps2i - zcp;
zs2f = zps2f - zcp;
ys3i = yps3i - ycp;
ys3f = yps3f - ycp;
zs3i = zps3i - zcp;
zs3f = zps3f - zcp;
(*Distances*)
ds1 = Simplify[ $\sqrt{(ys1f - ys1i)^2 + (zs1f - zs1i)^2}$ ];
ds2 = Simplify[ $\sqrt{(ys2f - ys2i)^2 + (zs2f - zs2i)^2}$ ];
ds3 = Simplify[ $\sqrt{(ys3f - ys3i)^2 + (zs3f - zs3i)^2}$ ];
(*Integrand Function*)
yls1[s1_] =  $\frac{ys1f - ys1i}{ds1} * s1 + ys1i$ ;
zls1[s1_] =  $\frac{zs1f - zs1i}{ds1} * s1 + zs1i$ ;

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$$y2s2[s2_]=\frac{ys2f-ys2i}{ds2}*s2+ys2i;$$

$$z2s2[s2_]=\frac{zs2f-zs2i}{ds2}*s2+zs2i;$$

$$y3s3[s3_]=\frac{ys3f-ys3i}{ds3}*s3+ys3i;$$

$$z3s3[s3_]=\frac{zs3f-zs3i}{ds3}*s3+zs3i;$$

(*First Area Moment at Sbar*)

$$Qz1[s1b_]=t*Integrate[y1s1[s1],\{s1,0,s1b\}];$$

$$Qy1[s1b_]=t*Integrate[z1s1[s1],\{s1,0,s1b\}];$$

$$Qz2[s2b_]=t*Integrate[y2s2[s2],\{s2,0,s2b\}];$$

$$Qy2[s2b_]=t*Integrate[z2s2[s2],\{s2,0,s2b\}];$$

$$Qz3[s3b_]=t*Integrate[y3s3[s3],\{s3,0,s3b\}];$$

$$Qy3[s3b_]=t*Integrate[z3s3[s3],\{s3,0,s3b\}];$$

(*Bottom Right S1*)

$$qs1vy0=0;$$

$$qs1vz0=0;$$

$$qs1vy[s1b_]=\frac{Vy*Iyz}{Iyy*Izz-Iyz^2}*Qy1[s1b]-\frac{Vy*Iyy}{Iyy*Izz-Iyz^2}*Qz1[s1b]+qs1vy0//Simplify;$$

$$qs1vz[s1b_]=-\frac{Vz*Izz}{Iyy*Izz-Iyz^2}*Qy1[s1b]+\frac{Vz*Iyz}{Iyy*Izz-Iyz^2}*Qz1[s1b]+qs1vz0//Simplify;$$

(*Middle Section S2*)

$$qs2vy0=qs1vy[h]-\frac{(-Vy*Iyz*zs2)+(Vy*Iyy*ys2)}{Iyy*Izz-Iyz^2}*As2;$$

$$qs2vz0=qs1vz[h]-\frac{(Vz*Izz*zs2)-(Vz*Iyz*ys2)}{Iyy*Izz-Iyz^2}*As2;$$

$$qs2vy[s2b_]=\frac{Vy*Iyz}{Iyy*Izz-Iyz^2}*Qy2[s2b]-\frac{Vy*Iyy}{Iyy*Izz-Iyz^2}*Qz2[s2b]+qs2vy0//Simplify;$$

$$qs2vz[s2b_]=-\frac{Vz*Izz}{Iyy*Izz-Iyz^2}*Qy2[s2b]+\frac{Vz*Iyz}{Iyy*Izz-Iyz^2}*Qz2[s2b]+qs2vz0//Simplify;$$

(*Top Left S3*)

$$qs3vy0=0;$$

$$qs3vz0=0;$$

$$qs3vy[s3b_]=\frac{Vy*Iyz}{Iyy*Izz-Iyz^2}*Qy3[s3b]-\frac{Vy*Iyy}{Iyy*Izz-Iyz^2}*Qz3[s3b]+qs3vy0//Simplify;$$

$$qs3vz[s3b_]=-\frac{Vz*Izz}{Iyy*Izz-Iyz^2}*Qy3[s3b]+\frac{Vz*Iyz}{Iyy*Izz-Iyz^2}*Qz3[s3b]+qs3vz0//Simplify;$$

(*Sanity Check, Shear Flow At Junctions Should Equal Zero*)

$$\text{Simplify}\left[qs1vy[h] - qs2vy[0] - \frac{(-Vy * Iyz * zs2) + (Vy * Iyy * ys2)}{Iyy * Izz - Iyz^2} * As2\right]$$

$$\text{Simplify}\left[qs1vz[h] - qs2vz[0] - \frac{(Vz * Izz * zs2) - (Vz * Iyz * ys2)}{Iyy * Izz - Iyz^2} * As2\right]$$

$$\text{Simplify}\left[qs2vy[h] + qs3vy\left[\frac{h}{2}\right] - \frac{(-Vy * Iyz * zs1) + (Vy * Iyy * ys1)}{Iyy * Izz - Iyz^2} * As1\right]$$

$$\text{Simplify}\left[qs2vz[h] + qs3vz\left[\frac{h}{2}\right] - \frac{(Vz * Izz * zs1) - (Vz * Iyz * ys1)}{Iyy * Izz - Iyz^2} * As1\right]$$

(*Find Branch Forces*)

F1vy = Integrate[qs1vy[s1b], {s1b, 0, h}];

F1vz = Integrate[qs1vz[s1b], {s1b, 0, h}];

F2vy = Integrate[qs2vy[s2b], {s2b, 0, h}];

F2vz = Integrate[qs2vz[s2b], {s2b, 0, h}];

F3vy = Integrate[qs3vy[s3b], {s3b, 0, $\frac{h}{2}$ }];

F3vz = Integrate[qs3vz[s3b], {s3b, 0, $\frac{h}{2}$ }];

(*Find Shear Center*)

Mxscvy = -F1vy * (ey - yc1) + F2vy * (ez - zc2) + F3vy * (ey - yc3);

Mxscvz = -F1vz * (ey - yc1) + F2vz * (ez - zc2) + F3vz * (ey - yc3);

eqneyez = Solve[{Mxscvy == 0, Mxscvz == 0}, {ey, ez}];

eys = ey /. eqneyez[[1]];

ezs = ez /. eqneyez[[1]];

ypey = eys + ycp;

zpey = ezs + zcp;

(*Questions 7-12*)

Q7 = qs1vz[0.0224] /. {Vy → 26, Vz → 15, h → 0.1083, t → 1.21 / 1000}

Q8 = qs2vz[0.0222] /. {Vy → 26, Vz → 15, h → 0.1397, t → 0.75 / 1000}

Q9 = F1vz /. {Vz → 130.19, h → 0.1304}

Q10 = F3vz /. {Vz → 12.98, h → 0.0996}

Q11 = (ypey /. {Vy → 41.78, Vz → 39.79, h → 0.1131, t → 1}) * 100

Q12 = (zpey /. {Vy → 126.38, Vz → 7.69, h → 0.0703}) * 100

Out[8]=

$$0. + 0.280172 h^3 t$$

Out[9]=

$$-\frac{1.57261 \times 10^{-16} Vy}{h}$$

Out[10]=

$$-\frac{2.55178 \times 10^{-16} Vz}{h}$$

$$Out[*]= \frac{0.601336 \text{ Vy}}{h} - \frac{0.601336 \text{ Vy}}{\sqrt{h^2}}$$

$$Out[*]= \frac{0.632517 \text{ Vz}}{h} - \frac{0.632517 \text{ Vz}}{\sqrt{h^2}}$$

$$Out[*]= 87.6063$$

$$Out[*]= 22.2598$$

$$Out[*]= 116.047$$

$$Out[*]= -1.4101$$

$$Out[*]= 1.22868$$

$$Out[*]= 6.46896$$